

This document contains the results generated by a Large Language Model (LLM) for selected phenotypes and traits. For each phenotype, the LLM analyzed the available biological and genetic context and identified the three most relevant cell types that are likely to contribute to the phenotype's underlying mechanisms. Along with each selected cell type, the model provides a biological rationale explaining its relevance to the phenotype.

Atrial fibrillation(30061737)

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[
  {
    "RANK 1 ": "Adult A Cardiomyocyte",
    "REASON ": "Atrial fibrillation is primarily a pathology of the atrial myocardium. Adult atrial cardiomyocytes host the critical electrophysiological and structural substrates (such as ion channels and gap junctions) whose genetic variations drive disease susceptibility."
  },
  {
    "RANK 2 ": "Fetal A Cardiomyocyte",
    "REASON ": "Many GWAS risk loci for atrial fibrillation are localized within developmental enhancers active during cardiogenesis, making fetal atrial cardiomyocytes essential for capturing the early transcriptional programs that predispose individuals to the arrhythmia."
  },
  {
    "RANK 3 ": "Adult V Cardiomyocyte",
    "REASON ": "While the phenotype is localized to the atria, atrial and ventricular cardiomyocytes share highly conserved cardiac-specific gene regulatory networks and contractile machinery, leading to significant heritability enrichment in the ventricular compartment."
  }
]
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Coronary Artery Disease(28714975)

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[
  {
    "RANK 1 "Adult Vasc Sm Muscle",
    "REASON": "Vascular smooth muscle cells (SMCs) are central to the structural integrity of coronary arteries. In coronary artery disease, SMCs undergo phenotypic switching, proliferation, and migration to form the fibrous cap of atherosclerotic plaques, making their regulatory genomic landscape highly enriched for CAD heritability."
  },
  {
    "RANK 2 ": "Adult Vasc Endothelial",
    "reason": "Endothelial cells line the vascular lumen and regulate vascular tone, inflammation, and barrier function. Endothelial dysfunction is a primary initiating event in atherogenesis, and genetic variants associated with CAD significantly alter endothelial enhancers and transcription factor binding sites."
  },
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{
  "RANK 3 ": "Fetal Vasc Sm Muscle",
  "reason": "Epigenetic programs active during fetal vascular development are frequently reactivated during pathological vascular remodeling and phenotypic modulation in adult coronary artery disease. CAD risk variants are strongly enriched in chromatin-accessible regions of developmental vascular lineages."
}
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]

Allergy eczema (UKB)

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[
  {
    "RANK 1 ": "Adult Bld CD4+ T Cell",
    "reason": "Atopic dermatitis and eczema are classically driven by dysregulated Type 2 (Th2) helper T cell responses. CD4+ T cells in the adult peripheral blood compartment show highly significant heritability enrichment for allergy and eczema risk variants."
  },
  {
    "RANK 2 ": "Fetal Thym T Cell",
    "reason": "The thymus is the primary site for T cell ontogeny and selection. Genetic variants associated with eczema and atopic susceptibility act early during development to alter fetal thymic T cell differentiation and central tolerance mechanisms."
  },
  {
    "RANK 3 ": "Adult Spl T Cell",
    "reason": "As a major secondary lymphoid organ, the spleen coordinates systemic immune responses. Adult splenic T cells harbor significant enrichment for eczema GWAS signals, reflecting their role in peripheral immune activation and memory cell generation in chronic inflammatory skin states."
  }
]
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Major depressive disorder(30718901)

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[
  {
    "RANK 1 ": "Adult Ctx Excitatory Neuron",
    "reason": "GWAS for Major Depressive Disorder (MDD) consistently shows strong heritability enrichment in the adult prefrontal cortex, specifically within glutamatergic excitatory neurons that govern synaptic plasticity, network connectivity, and mood regulation."
  },
  {
    "RANK 2 ": "Adult Ctx Inhibitory Neuron",
    "reason": "GABAergic inhibitory interneurons in the adult cortex are central to the excitation-inhibition imbalance hypothesis of depression and represent a major site of genomic enrichment for psychiatric trait susceptibility."
  },
  {
    "RANK 3 ": "Adult Ctx Excitatory Neuron",
    "reason": "GWAS for Major Depressive Disorder (MDD) consistently shows strong heritability enrichment in the adult prefrontal cortex, specifically within glutamatergic excitatory neurons that govern synaptic plasticity, network connectivity, and mood regulation."
  }
]
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{
  "RANK 3 ": "Fetal Ctx Excitatory Neuron",
  "reason": "MDD has a significant neurodevelopmental component; heritability
enrichment is prominent in fetal cortical excitatory neurons, indicating that genetic
risk factors impact early cortical neurogenesis and pioneering axonal projections."
}
]
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Thyroid stimulating hormone(30367059)

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[
  {
    "RANK 1 ": "Adult Thy Epithelial",
    "reason": "Thyroid-stimulating hormone (TSH) directly targets thyroid follicular
epithelial cells via the TSH receptor (TSHR) to regulate thyroid hormone synthesis and
secretion. Genetic variants associated with circulating TSH levels show strong
heritability enrichment in active chromatin regions of these adult thyrocytes."
  },
  {
    "RANK 2 ": "Fetal Thy Epithelial",
    "reason": "The developmental origins of the thyroid gland and the establishment of
the hypothalamic-pituitary-thyroid (HPT) axis setpoint are regulated by gene
expression program dynamics in fetal thyroid epithelial cells, making them highly
relevant to functional genomic variation in TSH levels."
  },
  {
    "RANK 3 ": "Adult Pit Endocrine",
    "reason": "TSH is synthesized and secreted by the thyrotropic subset of endocrine
cells within the anterior pituitary gland. Regulatory elements in these endocrine
cells control the expression and release of TSH in response to hypothalamic
thyrotropin-releasing hormone (TRH) and thyroid hormone feedback."
  }
]
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TYPE 2 diabetes(30297969)

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[
  {
    "RANK 1 ": "Adult Pan Beta cell",
    "reason": "Adult pancreatic beta cells are the primary site of insulin production
and secretion. Regulatory elements active in these cells show the most pronounced
enrichment for Type 2 diabetes heritability, driven by risk variants that impair
insulin secretory capacity and cell identity."
  },
  {
    "RANK 2 ": "Fetal Pan Beta cell",
    "reason": "Regulatory programs during fetal pancreatic development dictate
beta-cell mass and initial functional capacity. T2D risk variants are heavily enriched
in chromatin accessible regions of fetal beta cells, indicating that developmental
predisposition plays a key role in the pathogenesis of the disease."
  },
  {
    "RANK 3 ": "Adult Pan Alpha cell",
    "reason": "Pancreatic alpha cells secrete glucagon and cooperate with beta cells
to maintain glucose homeostasis. Epigenomic profiling demonstrates significant T2D
GWAS heritability enrichment in alpha cell regulatory elements, reflecting the
contribution of glucagon dysregulation to the diabetic phenotype."
  }
]
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    }
  ]
  Heightz (UKB)
  [
    {
      "RANK 1 ": "Fetal Lim Chondrocyte",
      "REASON ": "Chondrocytes within the fetal limb growth plates are the primary
drivers of endochondral ossification, the physiological process responsible for
longitudinal bone growth and a major determinant of adult height variation."
    },
    {
      "RANK 2 ": "Fetal Skel Osteoblast",
      "REASON ": "Osteoblasts in the fetal skeletal compartment are responsible for bone
matrix deposition and mineralization during early development, laying down the
structural foundation that dictates final stature."
    },
    {
      "RANK 3 ": "Adult Vasc Sm Muscle",
      "REASON ": "Vascular smooth muscle cells show strong heritability enrichment for
height GWAS signals, reflecting the systemic role of vascular maintenance,
extracellular matrix remodeling, and connective tissue pathways in determining
physical growth."
    }
  ]

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